

Phase 8: Data Management & Deployment

Phase 8 of the SmartShop CRM project focuses on Data Management and Deployment to ensure clean, reliable data in production and a safe, repeatable rollout process for updates.

8.1 Data Management

Data Management in Salesforce covers importing, exporting, updating, and maintaining records to keep processes, automation, and reports accurate and trusted.

Tools and approach

- Primary tool: Data Loader was used for imports and updates because it supports large volumes, upsert keys, robust error files, and repeatable mappings.
- Secondary option: The Data Import Wizard remains available for small, guided loads, but SmartShop prioritized Data Loader for control and scalability.

Data preparation

1. CSV file creation
 - Product2.csv was prepared with columns matching API names for standard products: Name, ProductCode (SKU), Category__c, Stock_Quantity__c, Low_Stock_Alert_Level__c, and Description.
 - PricebookEntry.csv captured Standard Price Book pricing by mapping Product2 to UnitPrice with IsActive for governed prices.
2. Data cleansing
 - Duplicates were removed, data types validated (for example, numeric fields for stock and price), and any date fields standardized to YYYY-MM-DD.
 - Column lengths and picklist values were checked against field definitions to avoid load-time rejections.
3. Mapping and referential checks
 - Lookup dependencies were validated before import, including ensuring the Standard Price Book was active and ProductCode aligned to the intended PricebookEntry rows.

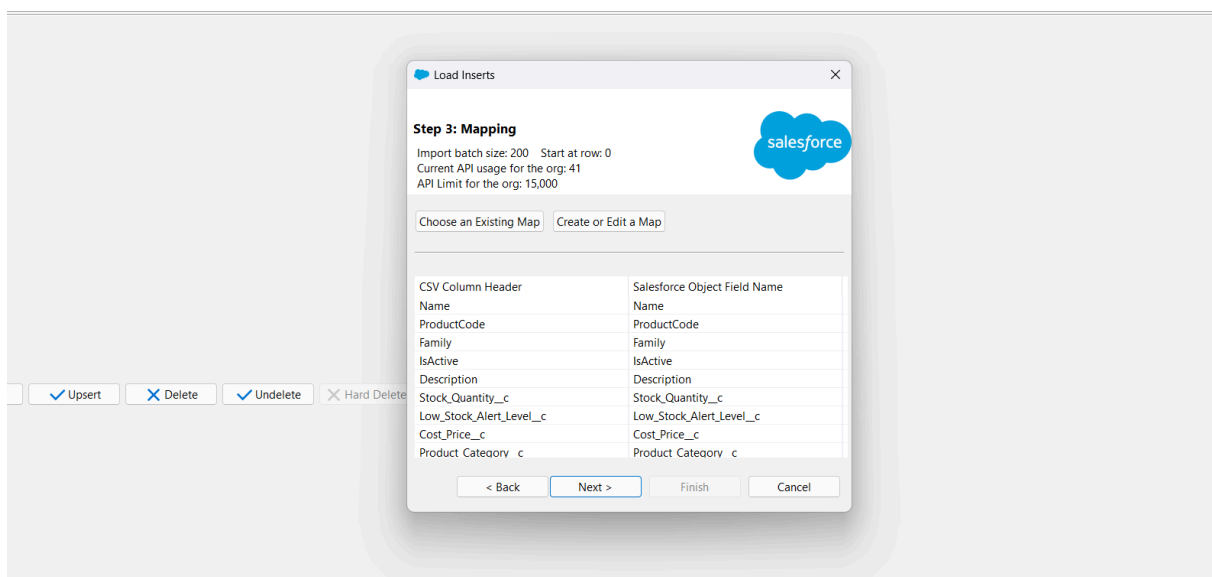
Data import execution (with Data Loader)

1. Load order
 - Insert Product2 records first to establish the catalog, then insert PricebookEntry rows for the Standard Price Book, followed by any Account and Contact seeds.

- Where applicable, Accounts were upserted using an External ID (Ext_Code__c) to make imports idempotent.
2. Mappings and run
- Data Loader maps CSV columns to Salesforce fields, saves mappings for reuse, and executes bulk inserts or upserts with clear success and error CSV outputs.
 - Bulk API or batch size adjustments were used as needed for performance and reliability.
3. Monitoring and reconciliation
- Success and error files were reviewed immediately, corrections were applied to source CSVs, and failed rows were reprocessed with the same External IDs to avoid duplicates.
 - Quick list views and starter reports (for example, Products by Category, Products without Pricebook Entries) confirmed referential integrity post-load.

Best practices implemented

- Validate before import: pre-check numeric formats, picklists, and dates in spreadsheets to reduce rejects
- Backup: run a data export before major loads to preserve a restorable snapshot.
- Incremental loads: import in small batches for easier troubleshooting and safer rollback.
- Error handling: correct and re-run only failed rows using consistent External IDs to keep data clean and auditable.



	A	B	C	D	E	F	G	H	I	J	K
1	Name	ProductCode	Family	IsActive	Description	Stock_Quantity	Low_Stock	Cost_Price	Product_Category__c		
2	iPhone 15	ELEC-IP15	Electronics	TRUE	Flagship smartphone	25	5	98000	Premium		
3	Blue Cotton Shirt	CLOT-BCS	Clothing	TRUE	100% cotton, button-down	50	10	500	Apparel		
4	Yoga Mat	SPRT-YMP	Sports	TRUE	Non-slip, eco-friendly	8	10	750	Fitness		
5	LED Desk Lamp	HOME-LED	Home & Garden	TRUE	Adjustable, touch control	4	10	1200	Home		
6	Noise Cancelling Headphones	ELEC-NCH	Electronics	TRUE	Over-ear, wireless	12	5	3200	Premium		
7	Stainless Steel Water Bottle	HOME-SSE	Home & Garden	TRUE	Insulated, leak-proof	60	15	350	Essentials		
8	Running Shoes	SPRT-RSP	Sports	TRUE	Lightweight, cushioned	22	8	1800	Fitness		
9	Casual T-shirt	CLOT-CTB	Clothing	TRUE	Unisex cotton	80	20	250	Apparel		
10	Bluetooth Speaker	ELEC-BSM	Electronics	TRUE	Portable, battery-powered	30	10	1500	Essentials		
11	Ceramic Coffee Mug	HOME-CCM	Home & Garden	TRUE	350ml capacity	120	25	120	Home		
12	Fitness Resistance Bands	SPRT-FRB	Sports	TRUE	Set of 5 bands	40	12	400	Fitness		
13	Formal Shirt	CLOT-FSW	Clothing	TRUE	Slim fit, wrinkle-free	35	10	600	Apparel		
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8.2 Deployment

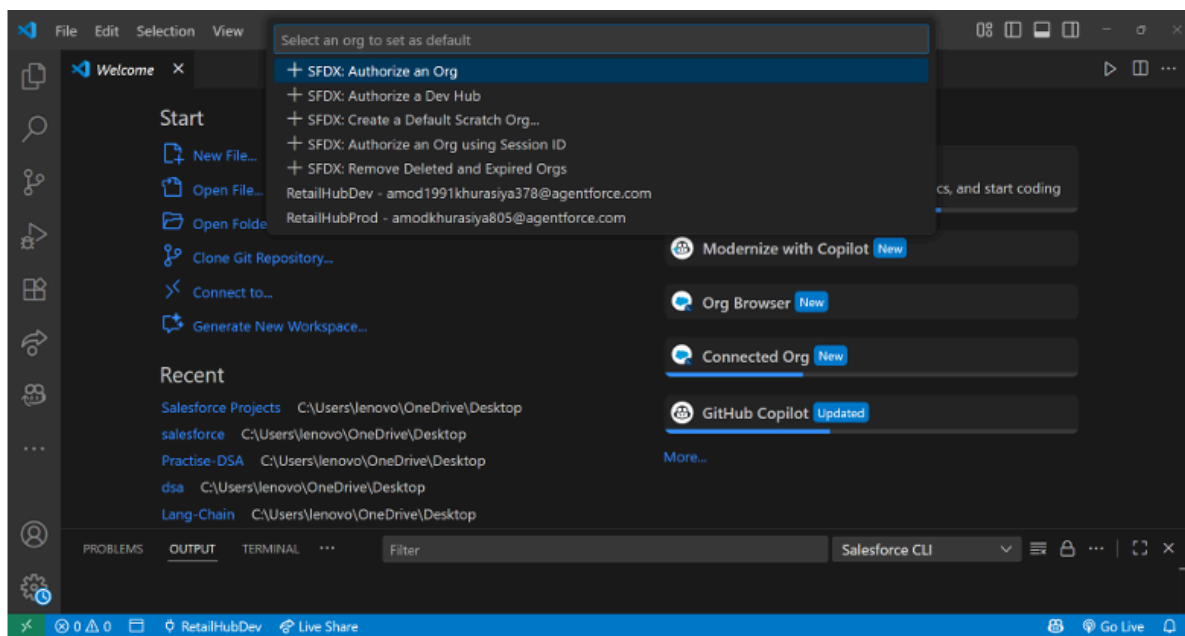
Deployment moves SmartShop metadata from the development org to production in a structured, low-risk way using VS Code and Salesforce DX (SFDX). This approach supports version control, repeatability, and clean handling of dependencies.

8.2.1 Local dev environment setup

- VS Code as the primary IDE for metadata work and source control.
- Salesforce Extension Pack for SFDX integration, syntax, and commands in the editor.
- Salesforce CLI (SFDX) installed and added to PATH for terminal commands and automation.
- Local project repository initialized to track changes and enable Git workflows.

8.2.2 Project creation and org authorization

- SFDX project initialized with a manifest (package.xml) to precisely define retrievable/deployable components.
- Authorized both source (development) and target (production) orgs via SFDX “Authorize an Org,” assigning easy aliases for commands.
- Retrieved metadata to local using “Retrieve Source from Org” against package.xml to keep a single source of truth.



8.2.3 Deployment method and waves

- Primary method: Deploy from VS Code using SFDX “Deploy This Source to Org” or CLI project deploy commands for non-source-tracked orgs.
- Multi-wave deployment to resolve dependencies: first deploy core schema (custom fields on Product2/Order/Account), then Flows and Validation Rules, then Apex (triggers/handlers/tests), followed by Lightning pages, reports, and dashboards.
- Profiles and permission sets updated alongside app assignments to prevent missing access after deploy.

8.2.4 Pre-deploy checks and validation

- Confirm Standard Price Book is active in target; verify component presence via package.xml.
- Validate the deployment by running a no-run/validation where supported, or deploy to a lower sandbox first for smoke checks.
- Ensure all Apex tests pass locally; plan to run “Run All Tests” in target during the final deploy to guarantee coverage and correct behavior.

8.2.5 Post-deploy activation and smoke tests

- Activate Flows in safe order: before-save record-triggered first, then after-save and scheduled; verify email deliverability settings.

- Smoke tests: create a Draft Order, use the “Add Products (Smart)” Screen Flow, verify OrderItem lines and Order total; optionally Activate to confirm inventory deduction and loyalty transaction creation by existing automations.
- Confirm dashboards and list views load with correct data and row-level visibility.

8.2.6 Rollback and audit

- Rollback strategy: if defects are found, deactivate new Flows, redeploy the previous trigger/class artifact from version control, and restore any impacted data from the latest export.
- Audit trail: keep SFDX command outputs, deployment logs, and test results; commit changes with clear messages and tag releases to reproduce deployments.

8.2.7 Benefits realized

- Professional, scalable deployments with VS Code, SFDX, and Git-based collaboration.
- Reduced risk via dependency-aware waves and pre-deploy validation; faster recovery thanks to a documented rollback path.